Software Maintenance, Refactoring & Evolution

RCS 305 ? Software Engineering | Riara University Revision Resource

1. The Four Categories of Software Maintenance (ISO/IEC 14764)

Maintenance represents 60% to 80% of total lifecycle software expenditure. It comprises:

? Corrective Maintenance: Reactive fixes resolving reactive bugs, security patches, and runtime crashes (approx. 20% of effort).

? Adaptive Maintenance: Modifying software to function across evolving operating systems, cloud runtimes, or hardware changes (approx. 25% of effort).

? Perfective Maintenance: Proactively implementing requested feature enhancements, performance optimizations, and user interface improvements (approx. 50% of effort).

? Preventive Maintenance: Restructuring, updating dependencies, and refactoring to prevent future failure modes (approx. 5% of effort).

2. Lehman’s Laws of Software Evolution

? Continuing Change: A system must continually adapt to its operational environment or become progressively less satisfactory.

? Increasing Complexity: As a system evolves, its complexity increases unless work is done to maintain or simplify structure.

3. Code Refactoring & Technical Debt

Refactoring alters internal structure without changing external observable behavior, clearing technical debt and restoring architectural cohesion.

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